

Table 1: One-pass residual readout vs. fairly-conditioned autoregressive loop (accuracy). Readout = small MLP on the frozen final residual (4-seed vote); loop = next-token log-prob over the closed answer set, full context, balanced few-shot. Δ = readout – loop.k, paired McNemar on identical rows (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). All runs batch 8; Qwen3 0.6B/4B on H200, Qwen3-8B/Mistral-7B/Granite-3.1-8B on B200, DeepSeek-V4-Flash on B300. ^a BoolQ loops: Qwen at pad 8192 with best $k \in \{0, \dots, 64\}$, Mistral/Granite/DSV4 at pad 2048, $k=8$. ^b RuleTaker n2k: readout scored on the loop’s exact 400 rows. ^c ARC-Challenge: full test (1165 rows), $k=8$, pad 2048.

Task	Model	Readout	loop.0	loop.k	Δ	p (McNemar)
BoolQ ^a	Qwen3-0.6B	0.753	0.631	0.715	+0.038***	2.4e−05
	Qwen3-4B	0.862	0.854	0.869	−0.007	0.22
	Qwen3-8B	0.879	0.862	0.886	−0.007	0.24
	Mistral-7B	0.841	0.798	0.852	−0.011	0.21
	Granite-3.1-8B	0.854	0.815	0.864	−0.010	0.64
	DeepSeek-V4-Flash	0.896	0.888	0.906	−0.009	0.39
RuleTaker n2k ^b	Qwen3-0.6B	0.638	0.600	0.545	+0.093**	3.7e−03
	Qwen3-4B	0.738	0.653	0.698	+0.040	0.20
	Qwen3-8B	0.753	0.675	0.730	+0.023	0.45
	Mistral-7B	0.660	0.515	0.575	+0.085*	0.017
	Granite-3.1-8B	0.828	0.558	0.695	+0.133***	1.3e−06
	DeepSeek-V4-Flash	0.763	0.605	0.838	−0.074**	1e−03
ARC-Challenge ^c	Qwen3-0.6B	0.492	0.502	0.597	−0.106***	3.2e−11
	Qwen3-4B	0.837	0.826	0.882	−0.046***	2.9e−06
	Qwen3-8B	0.908	0.902	0.913	−0.005	0.57
	Mistral-7B	0.728	0.750	0.779	−0.051***	9.3e−05
	Granite-3.1-8B	0.708	0.732	0.790	−0.082***	5.8e−10
	DeepSeek-V4-Flash	0.951	0.961	0.959	−0.007	0.23

Table 2: Readout probe-placement sweeps (18 cells, all batch 8): best mid-depth tap vs the final-layer readout, paired McNemar on identical rows. *Layers* = stack depth (28/36/36/32/40/43); taps are residual-layer indices. *Plateau onset* = first swept layer within 1 pt of the final-layer readout — the verdict representation is fully assembled from there on; at DeepSeek-V4 this happens by L22/42 (52% depth) on *all three tasks*, with ARC showing the sharpest snap-in (near-chance at L15 → 0.944 at L22). Mid-depth taps win significantly at 9/18 cells and never lose significantly.

Task	Model	Layers	Best tap	Depth	Plateau onset	mlp (best)	mlp (final)	Δ	p
BoolQ	Qwen3-0.6B	28	L18	64%	L18 (67%)	0.761	0.746	+0.010	0.18
	Qwen3-4B	36	L24	67%	L24 (69%)	0.868	0.856	+0.011*	0.038
	Qwen3-8B	36	L30	83%	L24 (69%)	0.889	0.878	+0.011*	0.016
	Mistral-7B	32	L16	50%	L16 (52%)	0.863	0.841	+0.022***	1.1e−04
	Granite-3.1-8B	40	L20	51%	L17 (44%)	0.868	0.854	+0.018**	1.0e−03
	DeepSeek-V4-Flash	43	L36	86%	L22 (52%)	0.900	0.896	+0.005	0.20
RuleTaker n2k	Qwen3-0.6B	28	L18	64%	L13 (48%)	0.688	0.650	+0.036*	0.020
	Qwen3-4B	36	L24	67%	L24 (69%)	0.781	0.744	+0.038**	2.8e−03
	Qwen3-8B	36	L24	67%	L24 (69%)	0.778	0.758	+0.025*	0.040
	Mistral-7B	32	L18	56%	L11 (35%)	0.702	0.643	+0.059***	5.2e−05
	Granite-3.1-8B	40	L20	51%	L20 (51%)	0.816	0.805	+0.011	0.37
	DeepSeek-V4-Flash	43	L29	67%	L22 (52%)	0.776	0.765	+0.021	0.057
ARC-Challenge	Qwen3-0.6B	28	L26	93%	L26 (96%)	0.500	0.492	+0.013	0.082
	Qwen3-4B	36	L29	81%	L24 (69%)	0.856	0.841	+0.022***	2.2e−04
	Qwen3-8B	36	L35	97%	L24 (69%)	0.909	0.909	+0.001	1.00
	Mistral-7B	32	L20	63%	L18 (58%)	0.741	0.728	+0.013	0.50
	Granite-3.1-8B	40	L39	100%	L39 (100%)	0.708	0.708	0.000	—
	DeepSeek-V4-Flash	43	L42	98%	L22 (52%)	0.951	0.951	−0.005	0.29

Table 3: What each access to the frozen residual costs (economics orientation, not a benchmark). Status stamps every row: “measured” = our run protocol; “projected” = forward arithmetic we did not run; “off-axis” = a real competing claim on a different cost axis (memory bandwidth), not operator FLOPs. Readout & scoring-loop are the same single pass at equal context — the readout’s structural gap is **no generation**, not layer-skipping. Early-exit serving shares our placement premise and is prior art for any “skip the tail” claim.

Access	Cost shape	Generates?	Status
readout (final tap)	1 fwd , short ctx	no	measured
scoring loop “.8”	1 fwd, long ctx	no	measured
early-exit decode	1 fwd, tail truncated	yes	projected / prior art
decode / output	K fwds (per token)	yes	measured (no self-stop)
Verity / bandwidth	1 fwd, memory-bound	yes	off-axis

Note. The decode row *is* measured: greedy decode from the Answer: prompt never self-terminates (25/25 hit the 300-token cap at 0.6B; 7/8 hit the 1000-token cap at DeepSeek-V4, `results/decode_baseline.json`).

Table 4: Controls, all 18 cells. Every headline number clears all four controls. *shuffl*: head refit on permuted labels sits at chance. *randproj*: a head on random Gaussian projections retains accuracy with perm $\approx$ max $\gg$ noise — the signal is *diffuse*. *ctx*: a mean-pooled full-context readout underperforms the final-token readout (last.mlp) — the verdict is computed into the final state, not read off the passage. Chance accuracy is ≈ 0.62 (BoolQ), 0.50 (RuleTaker), 0.27 (ARC).

Task	Model	shuffl (null)	randproj max	randproj perm	randproj noise	ctx.mlp	last.mlp
BoolQ	Qwen3-0.6B	0.533	0.734	0.725	0.601	0.667	0.746
	Qwen3-4B	0.542	0.861	0.859	0.604	0.717	0.856
	Qwen3-8B	0.528	0.874	0.874	0.607	0.739	0.878
	Mistral-7B	0.537	0.810	0.803	0.607	0.685	0.841
	Granite-3.1-8B	0.532	0.848	0.849	0.607	0.679	0.854
	DeepSeek-V4	0.522	0.889	0.891	0.602	0.678	0.896
RuleTaker	Qwen3-0.6B	0.511	0.635	0.635	0.489	0.517	0.650
	Qwen3-4B	0.474	0.718	0.726	0.505	0.586	0.744
	Qwen3-8B	0.488	0.736	0.745	0.496	0.575	0.758
	Mistral-7B	0.495	0.645	0.634	0.496	0.503	0.643
	Granite-3.1-8B	0.491	0.756	0.757	0.496	0.545	0.805
	DeepSeek-V4	0.500	0.726	0.716	0.506	0.518	0.765
ARC	Qwen3-0.6B	0.278	0.470	0.470	0.254	0.278	0.492
	Qwen3-4B	0.269	0.802	0.803	0.255	0.349	0.841
	Qwen3-8B	0.287	0.869	0.867	0.251	0.399	0.909
	Mistral-7B	0.259	0.679	0.678	0.251	0.277	0.728
	Granite-3.1-8B	0.284	0.598	0.588	0.251	0.279	0.708
	DeepSeek-V4	0.269	0.942	0.940	0.236	0.439	0.951